

Chapter 5: Mathematics Logic

Class 23: Practical / Activities (20 Minutes)

Topic: Mathematics Logic Quiz, Brain Teaser Challenge, Team Puzzle Competition, Oral Reasoning Assessment & Practical Revision

Activity 1: Mathematics Logic Quiz (5 Minutes)

Objective

To assess students' understanding of number patterns, logical reasoning, mental mathematics, visual puzzles, and problem-solving skills through an interactive quiz.

Materials Required

- Mathematics Quiz Cards
- Whiteboard & Marker
- Scoreboard
- Timer
- Student Answer Sheets

Instructions

1. Divide the class into **4–5 teams**.
2. Ask one question to each team in turn.
3. Give **30 seconds** to discuss and answer.
4. Award **1 point** for each correct answer.
5. Continue until all questions are completed.
6. Announce the winning team and review the correct answers.

Sample Quiz Questions

1. Complete the pattern: **5, 10, 15, __, 25**
2. Which number comes after **199**?
3. Which is greater: **78 or 87**?

4. What is 9×8 ?
5. What is $72 \div 9$?
6. Arrange in ascending order: **45, 18, 62, 29**
7. Name two even numbers.
8. What is a number pattern?
9. What is logical reasoning?
10. Name one strategy game.

Teacher Observation Checklist

- ☒ Answers correctly.
 - ☒ Thinks logically.
 - ☒ Participates actively.
 - ☒ Works well in a team.
 - ☒ Demonstrates mathematical confidence.
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Activity 2: Brain Teaser Challenge (5 Minutes)

Objective

To improve students' observation, creativity, reasoning, and critical thinking through fun brain teasers.

Materials Required

- Brain Teaser Cards
- Whiteboard
- Activity Worksheet
- Timer

Instructions

1. Present one brain teaser at a time.
2. Give students **30–45 seconds** to think.
3. Students answer individually or in teams.
4. Encourage students to explain their reasoning.

5. Discuss the correct answers with the class.

Brain Teaser Questions

1. What comes once in a minute, twice in a moment, but never in an hour?

Answer: The letter **M**

2. Which is heavier: **1 kg of cotton** or **1 kg of iron**?

Answer: Both weigh **1 kg**.

3. I have keys but cannot open doors. What am I?

Answer: A keyboard.

4. If today is Monday, what day comes after two days?

Answer: Wednesday.

5. A clock has three hands. Name them.

Answer: Hour hand, Minute hand, Second hand.

Teacher Observation Checklist

- ☒ Thinks carefully.
 - ☒ Gives logical answers.
 - ☒ Explains reasoning.
 - ☒ Participates confidently.
 - ☒ Demonstrates creativity.
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Activity 3: Team Puzzle Competition (5 Minutes)

Objective

To strengthen teamwork, communication, observation, and mathematical reasoning through puzzle-solving activities.

Materials Required

- Puzzle Worksheets
- Shape Cards
- Number Cards
- Whiteboard
- Timer

Instructions

1. Divide the class into **small teams**.
2. Give each team one puzzle worksheet.
3. Students work together to solve all puzzles.
4. The first team to complete all puzzles correctly earns the highest score.
5. Review all answers with the class.

Puzzle Activities

Puzzle 1 – Complete the Number Pattern

10, 20, 30, __, 50

Puzzle 2 – Find the Odd One Out

Apple – Mango – Banana – Bus

Puzzle 3 – Shape Pattern

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Puzzle 4 – Arrange in Descending Order

25, 80, 45, 60

Puzzle 5 – Mental Calculation

35 + 45 = _____

Teacher Observation Checklist

- ☒ Solves puzzles accurately.
 - ☒ Works cooperatively.
 - ☒ Applies logical thinking.
 - ☒ Finishes within time.
 - ☒ Communicates effectively.
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Activity 4: Oral Reasoning Assessment & Practical Revision (5 Minutes)

Objective

To evaluate students' understanding of mathematical concepts, logical reasoning, and problem-solving through oral questions and quick revision.

Materials Required

- Oral Question Cards
- Teacher Assessment Sheet
- Whiteboard
- Revision Flashcards

Instructions

1. Call students one by one.
2. Ask each student **3–5 oral questions**.

3. Encourage complete and confident answers.
4. Conduct a quick revision of important concepts from the chapter.
5. Provide positive feedback and improvement suggestions.

Sample Oral Questions

- What is a number pattern?
- Name five odd numbers.
- What is skip counting?
- Which number comes before **500**?
- What is **12×6** ?
- What is **$90 \div 10$** ?
- Why is logical reasoning important?
- What is Sudoku?
- Name one strategy game.
- Give one example of mathematics used in daily life.

Practical Revision Checklist

Students should be able to:

- Identify number patterns.
- Complete number sequences.
- Solve simple visual puzzles.
- Perform mental calculations.
- Compare numbers correctly.
- Solve real-life word problems.
- Apply logical reasoning.
- Participate confidently in mathematical discussions.

Teacher Observation Checklist

- ☒ Speaks confidently.
 - ☒ Answers correctly.
 - ☒ Demonstrates logical thinking.
 - ☒ Applies mathematical concepts.
 - ☒ Participates actively.
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Practical Time Distribution

Activity	Duration
Mathematics Logic Quiz	5 Minutes
Brain Teaser Challenge	5 Minutes
Team Puzzle Competition	5 Minutes
Oral Reasoning Assessment & Practical Revision	5 Minutes
Total Practical Time	20 Minutes

Expected Learning Outcomes

After completing these practical activities, students will be able to:

- Revise and apply all concepts learned in **Chapter 5: Mathematics Logic** with confidence.
- Solve **number patterns, logical reasoning questions, brain teasers, and puzzles** accurately.
- Perform **mental mathematics** quickly and explain their reasoning clearly.
- Demonstrate strong **observation, analytical thinking, and problem-solving skills** during quizzes and team challenges.
- Communicate mathematical ideas confidently through **oral assessments and collaborative activities**.
- Build confidence, teamwork, and critical thinking skills while preparing for the chapter-end and final assessments.